

Taher Batterywala

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Education

Vellore Institute of Technology
B.Tech in CSE (Specialization in AIML)

Bhopal, India
Sep 2023 – Jun 2027

S.I.C.A. Senior Secondary School
Senior Secondary Education (CBSE, Class I-XII)

Indore, India
Apr 2008 – Mar 2023

Technical Skills

Programming Languages	C++, Python, SQL
CS Fundamentals	Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Computer Networks
AI & Machine Learning	Deep Learning, Computer Vision, Generative AI, Large Language Models (LLMs), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG)

Experience

FOSSEE, IIT Bombay | Summer Intern | [GitHub](#) | [Certificate](#) *May 2026 – Jul 2026*

- Implemented a subprocess isolation layer utilizing Base64-serialized I/O pipes for a local AI Copilot, resolving C++ memory segmentation faults and eliminating UI thread latency (from 1.5s to near-zero).
- Integrated an offline RAG pipeline using Qwen2.5:3b and MiniCPM-V, tokenizing and indexing engineering manuals into 74 discrete ChromaDB vector database layers for local retrieval.
- Wrote a regex-driven garbage collection routine to eliminate file-system memory leaks from temporary clipboard images, and built a self-healing config script to resolve startup crashes on fresh installations.

Projects

Tirakot AI: Offline Voice-Activated Desktop Assistant | [GitHub](#)

Local AI & Systems Engineering Project | **Tech:** Python, PyTorch, Ollama, Whisper, CustomTkinter, asyncio

- Engineered a dual-thread AI assistant running 3 ML models (Qwen 2.5, Whisper STT, Edge-TTS) concurrently under a strictly managed 2.34 GB VRAM budget on consumer hardware.
- Built 8 real-time OS automation tools (web search, app launching, VS Code generation) triggered by LLM JSON tool-calling via zero-latency first-token stream interception.
- Implemented an EMA-calibrated noise floor and 3-frame VAD gating for wake-word detection, completely eliminating Whisper transcription hallucinations during ambient silence.

GenWealth: Financial Forecasting & NLP System | [GitHub](#)

Deep Learning & Quant Finance Project | **Tech:** PyTorch (LSTM), Scikit-learn, FinBERT, yfinance

- Architected a hybrid forecasting engine (LSTM + Random Forest) trained on 200 equities across 16 years of market data, achieving 55% directional accuracy on unseen assets.
- Processed 1.84M raw headlines and executed GPU-accelerated FinBERT inference to generate daily confidence-weighted sentiment signals for 1.65M+ financial news records.
- Designed a deterministic "Logic Injection Layer" to correct model Permabull bias, validated by a 7.2% confidence differential during live sentiment stress testing scenarios.

GamerSense: Behavioral Segmentation Engine | [GitHub](#)

Data Science & ML Engineering Project | **Tech:** Python, Scikit-learn, XGBoost, SMOTE, Flask, Pandas

- Engineered an end-to-end ML pipeline with dataclass configurations, deploying K-Means clustering to segment 3,024 mobile players into actionable behavioral personas.
- Benchmarked 7 algorithms utilizing SMOTE on a severe 38:6:1 imbalanced dataset, achieving 77.2% classification accuracy with an SVC to predict in-app spending segments.

Leadership & Extracurriculars

Google Developer Groups VIT Bhopal | Core Team Member | [Certificate](#)

Nov 2024 – Jun 2025

Achievements & Certifications

- Competitive Programming** – Solved 400+ Data Structures and Algorithms problems on various coding platforms. [Profile](#)